HOVERCRAFT B/C

Team Checklist – 2027

Rank: _____
Final Score: _____

Team Number: _____ School & Team Name: _____
Student Names: _____

IMPOUND

1. T F	2.a.	Team impounds one Vehicle (batteries stored separately), any spare parts and any papers/notes that will be used. (If team fails to impound, a 0.7 multiplier is applied to each run's distance score (DS), time score (TS), and Mass score (MS))

CHECK-IN AND SAFETY

T F	2.c.	**Eye Protection:** Participants wear Class B eye protection during vehicle setup and operation.
T F	3.f.i.	**Propeller Shielding:** All propellers/impellers have shielding that prevents a ⅜" dowel from touching them.
T F	3.f.ii.	**Batteries:** The vehicle is powered by a maximum of 12.0V commercial batteries, including rechargeables, for each motor. The battery's label is the manufacturer's original and easily viewable by the ES. The expected voltage across any two points does not exceed 12.0V if multiple batteries are used. The vehicle has no other energy sources and no lithium or lead batteries.
T F	3.f.iii.	**Switch:** All motors have an accessible, manually-operated, mechanical switch to start and stop the vehicle and does not rely on inserting batteries or twisting wires to start the vehicle.
2. T F		**ALL SAFETY PARAMETERS ARE MET.** If F, do not run and team gets Participation points.

Fill out Box 3A with # of Construction Violations if no runs occur.

1st Run	2nd Run	3rd Run	EVENT TIME
Competition Time ends after two complete runs or two incomplete runs, or competition time limit reached (whichever comes first)			
			Construction Parameters:
T F	T F	T F	**3.a. Size:** Vehicle fits within a 40.0 cm x 40.0 cm x 40.0 cm box when levitated in the ready-to-run configuration and does not modify or damage the track.
T F	T F	T F	**3.b. Electrical Components:** are limited to batteries, wires (including battery holders/connectors), mechanical switches, relays, resistors (including potentiometers and rheostats), capacitors, and up to two motors (including brushless motors). Integrated circuits other than those that are an integrated manufactured part of a commercial motor are not used. Components with lasers are not used.
T F	T F	T F	**3.c Rails:** Vehicle components do not contact the top or outside edge of the side rails.
T F	T F	T F	**3.d. Thrust and Lift:** Vehicle levitates on a cushion of air as it moves down the track. All thrust and lift on the vehicle is achieved with air pressure.
T F	T F	T F	**3.e. Dowel:** The vehicle has a ¼" or larger wooden dowel vertically attached within 3.0cm of its front edge such that the top end is at least 20.0 cm above the track's surface when levitated. The dowel must be the first part of the vehicle to break a laser timing beam when the vehicle is travelling forward.
T F	T F	T F	**3.g. Knowledge:** Participants are able to answer questions regarding the design, construction, and operation of the vehicle per the Building Policy found on www.soinc.org.
T F	T F	T F	**4.j. Falling Off:** No part of the vehicle, nickel rolls, or individual nickels fall off during the run.
3A.	**3B.**	**3C.**	**# of unfixed Construction Parameter violations in this run (Each missed = 0.8x multiplier for DS, TS, and MS for that run) (If blank, run complies with "Construction Parameters")**
			Competition Parameters:
T F	T F	T F	**4.a. Outside Assistance:** Participants do not leave or receive outside assistance, materials, or communication from when they enter the event area to compete until they are finished competing.
T F	T F	T F	**4.l. Start:** Team places their vehicle, including the nickel load, on the track fully behind the start line. The team then activates their vehicle's motor(s) and gives a countdown of "3, 2, 1, launch".
T F	T F	T F	**4.m. Dowel:** If photogates are used, the dowel is the first part of the vehicle to break the laser beams at both the start and finish lines.
T F	T F	T F	**4.e. Practice Runs:** Team does not do any practice runs.
T F	T F	T F	**4.f. Track Surface:** Team does not wet or apply any substance to the track.
4A.	**4B.**	**4C.**	**# of Competition Parameter violations in this run (Each missed = 0.9x multiplier for DS, TS, MS for that run) (If blank, run complies with "Competition Parameters")**
5A. T F	**5B. T F**	**5C. T F**	**Ramp Not Used:** The Ramp is not used. If F, Distance Score is multiplied by 0.5 for that run.
6A. T F	**6B. T F**	**6C. T F**	**4.n. Interference:** After the vehicle starts, the team does not touch the vehicle or interfere with its run until the ES says so. If F, the run is counted as a complete run with a DS, TS, and MS of 0.
7A. C I	**7B. C I**	**7C. C I**	**Complete/Incomplete:** A complete run is where the Vehicle doesn't stop for more than 3 seconds before crossing the finish line and crosses the finish line within (2 x TT) seconds.
8A.	**8B.**	**8C.**	**Mass:** # of rolls of nickels that stay on the vehicle for the entirety of the run. A half roll counts as 0.5.
9A.	**9B.**	**9C.**	**Mass:** # of individual nickels that stay on the vehicle for the entirety of the run.
10A.	**10B.**	**10C.**	**Distance:** Dowel to finish line distance (cm). Put 0 for Distance for complete runs.
11A.	**11B.**	**11C.**	**Run Time:** Time (in s) from when the dowel crosses the start line to when the dowel crosses the finish. Leave blank for incomplete runs.

12. T F General Rule: The team is disqualified. (Notify the team and their coach as soon as possible.)

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